

Mobile Robotics - Assignment 2

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For all mobile robots, it is indispensable to have low-level control systems for the actuators. In the case of the differential robot used in the course, in order to utilize it for complex tasks, a velocity controller for each of the motors is needed. In this assignment, the formulation of a control system for the motor used will be presented. It will be assumed that the motors share the same dynamical model, and considerations on the kinematic model will be left to the next assignment.

1 Proportional-Integral Velocity Controller

In the previous assignment, the motor's dynamical model was approximated as a first-order system. This design choice will be relevant to the controller synthesis. First, consider the approximated transfer function:

$$G(s) = \frac{W(s)}{U(s)} = \frac{A}{\tau \cdot s + 1} = \frac{3.595}{0.1596 \cdot s + 1}, \quad (1)$$

Where $A = 3.595$ is the gain and $\tau = 0.1596$ is the system's time constant. This Single-Input, Single Output (SISO) system can be controlled via a linear feedback controller, provided that the closed-loop system is stable. As the open-loop system is stable with one pole and no zeros, tracking of step reference signal $R(s) = \frac{1}{s}$ can be done without steady-state error with a Proportional-Integral (PI) Controller. Consider the controller architecture presented in Figure 1.

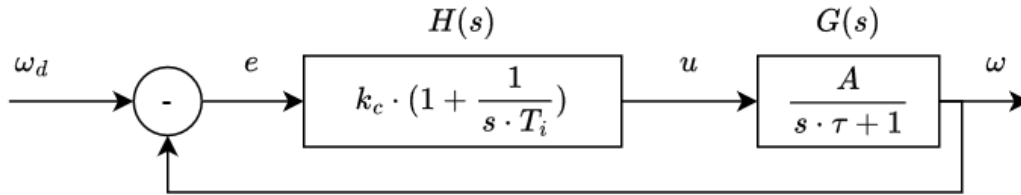


Figure 1: Closed-Loop system - Block diagram

The controller $H(s)$ is described by the transfer function:

$$H(s) = k_c \cdot \left(1 + \frac{1}{s \cdot T_i} \right) = k_c \frac{T_i \cdot s + 1}{s \cdot T_i}, \quad (2)$$

where k_c is the proportional gain and T_i is the integral time constant. The first design choice of the controller is to choose $T_i = \tau$, making the zero of the controller cancel the pole of the motor:

$$G(s) \cdot H(s) = k_c \frac{\cancel{s \cdot T_i + 1}}{s \cdot T_i} \frac{A}{\cancel{\tau \cdot s + 1}} = \frac{k_c \cdot A}{s \cdot \tau} \quad (3)$$

Therefore, the closed-loop transfer can be derived as:

$$\frac{G(s) \cdot H(s)}{1 + G(s) \cdot H(s)} = \frac{A \cdot k_c (T_i s + 1)}{T_i \tau s^2 + T_i s (1 + A \cdot k_c) + A \cdot k_c} = \frac{1}{s \cdot \frac{\tau}{k_c \cdot A} + 1} \quad (4)$$

This results once again in a first-order system, this time with $A_{cl} = 1$, as the system tracks any input. The only remaining parameter to choose is the closed-loop time constant τ_{cl} , as this value will

dictate how fast the control system can respond to changes in the reference signal. Then, the time constant of the closed-loop system is equal to:

$$\tau_{cl} = \frac{\tau}{k_c \cdot A}. \quad (5)$$

To guarantee that the system can track the reference signal without overshoot, the controller will be conservatively tuned by setting:

$$\tau_{cl} = 2 \cdot \tau \implies k_c = 0.1693 \quad (6)$$

The closed-loop system was subjected to reference of $w_d = 4 \text{ rad/s}$ to validate the controller synthesis. The step response to this reference signal was analyzed and can be visualized in Figure 2. From the figure, it can be empirically assessed that the value of the closed-loop time constant is $\tau_{cl} = 0.7 - 0.36 = 0.34$, which is similar to the designed value of 0.32 s , thus validating the controller design.

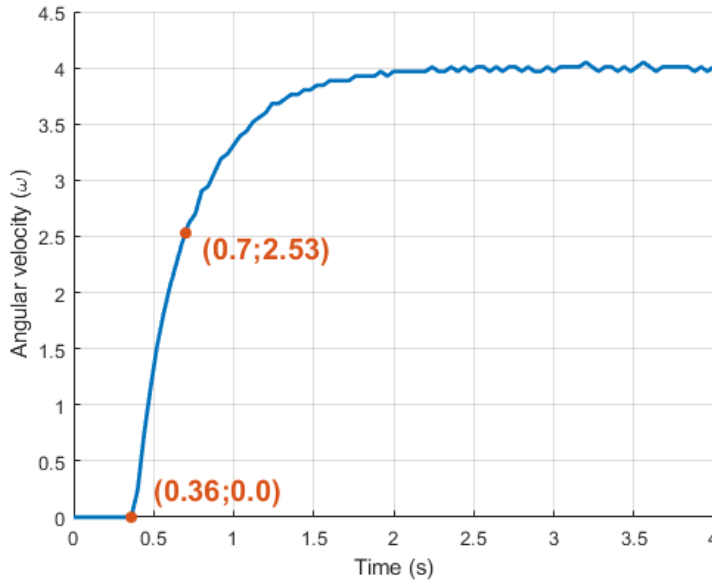


Figure 2: Closed-loop step response

2 Feedforward Signal

The previous section shows the design and synthesis of a linear feedback controller. While this controller is successful at tracking a reference signal, its convergence is relatively slow due to the conservative nature of the design choice $\tau_{cl} = 2\tau$. This section will describe the introduction of a feedforward signal to the control actuation, which will significantly speed up the closed-loop system.

Consider once again the system transfer function $G(s)$. Assuming this model perfectly depicts the system dynamics, then the theoretical best open-loop controller is to invert the plant with a feedforward controller $F(s) = \frac{1}{G(s)}$, resulting in the overall control system being:

$$F(s)G(s) = 1 \quad (7)$$

This means that no matter what the input signal is, the output will be exactly the same. Consider now the inversion of the motor plant:

$$F(s) = G(s) = \frac{s \cdot \tau}{A} + \frac{1}{A} = k_f + k_d \cdot s \quad (8)$$

This transfer function has a very glaring issue. It is not proper, meaning that it has more zeros than poles. This breaks the causality assumption of LTI systems and, therefore, makes in practice

the inversion of the plant infeasible without adding another pole in the form of a low-pass filter. Furthermore, even if feasible, considering that most of the references would be steps, their derivatives would be dirac impulses, meaning that the actuation signal would need infinite bandwidth, which is not realizable. For these two reasons and to keep the implementation simple, the k_d parameter will be considered zero for the remainder of the document.

The remaining parameter, k_f , will be equal to the inverse of the open-loop gain, $k_f = \frac{1}{A}$, which would be a naive approach for open-loop control. Due to the uncertainty on this parameter and in the transfer function as a whole (as it was approximated), the previous PI controller must also be introduced in the control architecture in order to make the necessary corrections to achieve no steady-state error. The full architecture can be visualized in [Figure 3](#)

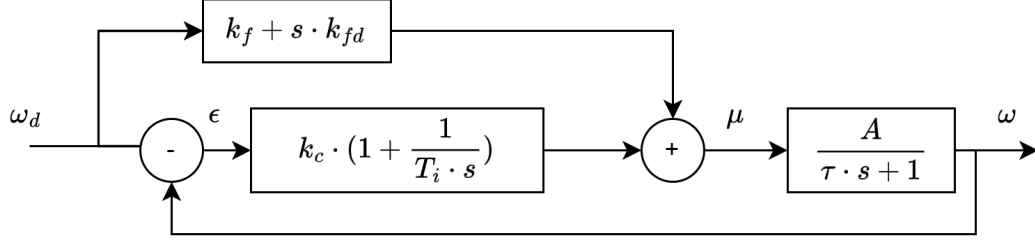


Figure 3: Closed-loop control architecture with feedforward - Block diagram

To evaluate the control system performance, multiple values of k_f were tested in the same conditions as in the previous section. Considering that the baseline value is $k_f = 0.34$, the other values will be 60%, 80% of this baseline. The corresponding step responses can be visualized in [Figure 4](#).

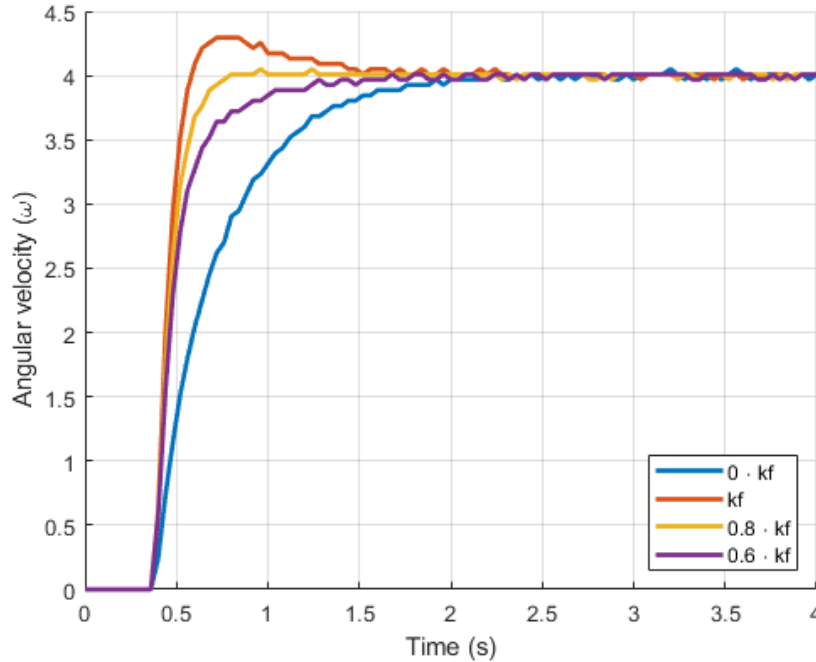


Figure 4: Step responses with multiple k_f values

From the figure, one can conclude that the best value for k_f , in this case, is $0.27 \approx k_f \cdot 80\%$, since it is the value for which the system has a faster response (compared with $k_f = 0 \vee 0.20$) and remains without overshoot, as is the case of $k_f = 0.34$. Table 1 contains the settling time (the time at which the output is at 2% of its steady-state value) and overshoot for all cases.

It is important to note that the system overshoots the reference with $k_f = 0.34$ because of actuation saturation, and in the case without these constraints, this value would most likely perform better.

Table 1: System performance evaluation

k_f (%)	Value	Settling Time 2% (s)	Overshoot (%)	Time Constant (s)
0	0	1.4	0	0.34
100	0.34	1.08	7.37	0.10
80	0.27	0.36	0	0.12
60	0.20	0.88	0	0.14

Furthermore, it is important to consider that the optimal value might depend on the magnitude of the step reference. If the step is smaller, the actuation is less likely to saturate, allowing higher values of k_f to perform better. A similar argument can be made to the inverse case where the step is larger and k_f smaller.

Another interesting observation from Table 1 and Figure 4 is that in all control schemes with feedforward, the response is very similar until $t = 0.1s$, as well as having similar time constants, despite having largely different settling times. This is due once again to the actuator saturation effect, as the system is responding as fast as possible. Once the velocity error starts going down, the actuator is no longer saturating, resulting in the divergence of responses. This effect clearly shows the difference between exponential convergence and finite-time convergence, the latter being exhibited by the feedforward controllers up until the proportional control starts reducing the actuation effort. Thus, all feedforward trajectories reach in finite time the set $\Omega_e = \{|e| \leq \epsilon\}$, where $kp \cdot |e| + k_f \cdot |e| = |u_{max}|$, considering that the integral error is approximately zero.

Thus, the choice of the k_f value can be seen as minimizing ϵ while ensuring the system does not overshoot. This response would be very similar to the reaching phase of the Sliding Mode Controller by approximating its finite-time convergence properties.

3 Conclusion

This document presented the design and analysis of two control strategies for a mobile robot's motor velocity control: a Proportional-Integral (PI) controller and a feedforward signal-based controller.

The PI controller, designed to eliminate steady-state error, demonstrates stable and accurate tracking of reference signals, as validated through empirical testing. However, its conservative design choice results in relatively slow convergence.

The introduction of a feedforward signal significantly enhances the closed-loop system's speed by inverting the open-loop gain of the motor's plant. Despite the practical challenges, such as actuation saturation, the feedforward controller offers faster response times compared to the PI controller. Through empirical analysis, an optimal feedforward gain (k_f) is determined to balance convergence speed and overshoot, showcasing its effectiveness in real-world applications.